Temperatures are obtained using equations of state:

$$E = E(p, T) \tag{2.18}$$

$$e_e = \frac{3}{2} k_B T_e \tag{2.19}$$

$$e_i = e_i(p_i, T_i) \tag{2.20}$$

2.7 The complete system of equations

The system of equations have a total of 21 nonlinear variables, organized into four categories. Five equations are needed to describe the fluid motion of neutral particles.

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \rho \mathbf{u} = -\frac{M}{N_A}(r_{\mathrm{pi}} + r_{\mathrm{ci}} - r_{\mathrm{rec}}) \tag{2.21}$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot \rho \mathbf{u} \otimes \mathbf{u} = -\nabla p + \nabla \cdot \mathbf{T} + \rho \mathbf{g} + \frac{M}{N_A}(\mathbf{f}_{e \to n} + \mathbf{f}_{i \to n}) \tag{2.22}$$

$$\frac{\partial \rho E}{\partial t} + \nabla \cdot \rho \mathbf{u} H = -\nabla \cdot \mathbf{q}'' + \nabla \cdot [\mathbf{T} \cdot \mathbf{u}] + q'''_{e \to n} + q'''_{i \to n} \tag{2.23}$$

Electron motion requires another set of five equations.

$$\frac{\partial N_e}{\partial t} + \nabla \cdot N_e \mathbf{u}_e = r_{\mathrm{pi}} + r_{\mathrm{ci}} - r_{\mathrm{rec}} \tag{2.24}$$

$$\frac{\partial N_e \mathbf{u}_e}{\partial t} + \nabla \cdot N_e \mathbf{u}_e \otimes \mathbf{u}_e = \frac{-\nabla p_e - e N_e (\mathbf{E} + \mathbf{u}_e \times \mathbf{B})}{m_e} - \mathbf{f}_{e \to n} \tag{2.25}$$

$$\frac{\partial N_e e_e}{\partial t} + \nabla \cdot N_e \mathbf{u}_e H_e = q'''_{\mathrm{laser}} - q'''_{e \to n} - q'''_{e \to i} - e N_e \mathbf{u}_e \cdot \mathbf{E} \tag{2.26}$$

Ion motion is described by a very similar set of equations.

$$\frac{\partial N_i}{\partial t} + \nabla \cdot N_i \mathbf{u}_i = r_{\mathrm{pi}} + r_{\mathrm{ci}} - r_{\mathrm{rec}} \tag{2.27}$$

$$\frac{\partial N_i \mathbf{u}_i}{\partial t} + \nabla \cdot N_i \mathbf{u}_i \otimes \mathbf{u}_i = \frac{-\nabla p_i + e N_i (\mathbf{E} + \mathbf{u}_i \times \mathbf{B})}{m_i} - \mathbf{f}_{i \to n} \tag{2.28}$$

$$\frac{\partial N_i e_i}{\partial t} + \nabla \cdot N_i \mathbf{u}_i H_i = q'''_{e \to i} - q'''_{i \to n} + e N_i \mathbf{u}_i \cdot \mathbf{E} \tag{2.29}$$

Finally, the electric and magnetic fields that arise from charge movement are taken from Maxwell's Equations.

$$\varepsilon_r \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} = \frac{1}{\mu_0} \nabla \times \mathbf{B} - e(N_i \mathbf{u}_i - N_e \mathbf{u}_e) \tag{2.30}$$

$$\frac{\partial \mathbf{B}}{\partial t} = -\nabla \times \mathbf{E} \tag{2.31}$$